EC2 roles and URL fetching

AWS Academy is the main lesson. This short deck connects it to our class work.

PRIMARY

AWS Academy

Cloud Foundations · Module 6 EC2 portion

IN CLASS

Practice

Lab 3 · Introduction to Amazon EC2

COURSE WORK

Local task

Local 02 starts · C5R1 Part 1 / C5R2 Part 2

Local 02 due 30 Nov · Use only your assigned Part

One path through the topic

STEP 1

Instance role

An EC2 app can receive temporary credentials without a key in its code.

STEP 2

Unsafe fetch

A server that fetches any user URL can be asked to reach its own metadata path.

STEP 3

Control check

Reject internal targets and confirm an intended fetch still works.

Why is blocking only the literal metadata IP insufficient?

What to do this week

BEFORE CLASS

Prepare

Open the EC2 slides and prepare Docker if you are assigned Part 2.

IN CLASS

Work

Complete the EC2 lab, then compare vulnerable and fixed local URL fetching.

AFTER CLASS

Record

Submit the assigned Part in Local 02. Keep credentials out of the sheet.

AWS work stays in AWS Academy · Local work stays in Classroom

Instance role and unsafe fetch

METADATA

On the instance

The EC2 metadata endpoint is reachable from the instance; an outside caller cannot fetch it directly.

SSRF

Borrowed access

An attacker-controlled URL makes the app fetch an internal target using the instance's network access.

FIX

Check destinations

Allow intended hosts; validate resolved addresses and redirects before each request. Keep the role least-privileged.

Local 02: a login check alone does not make an unsafe URL safe.